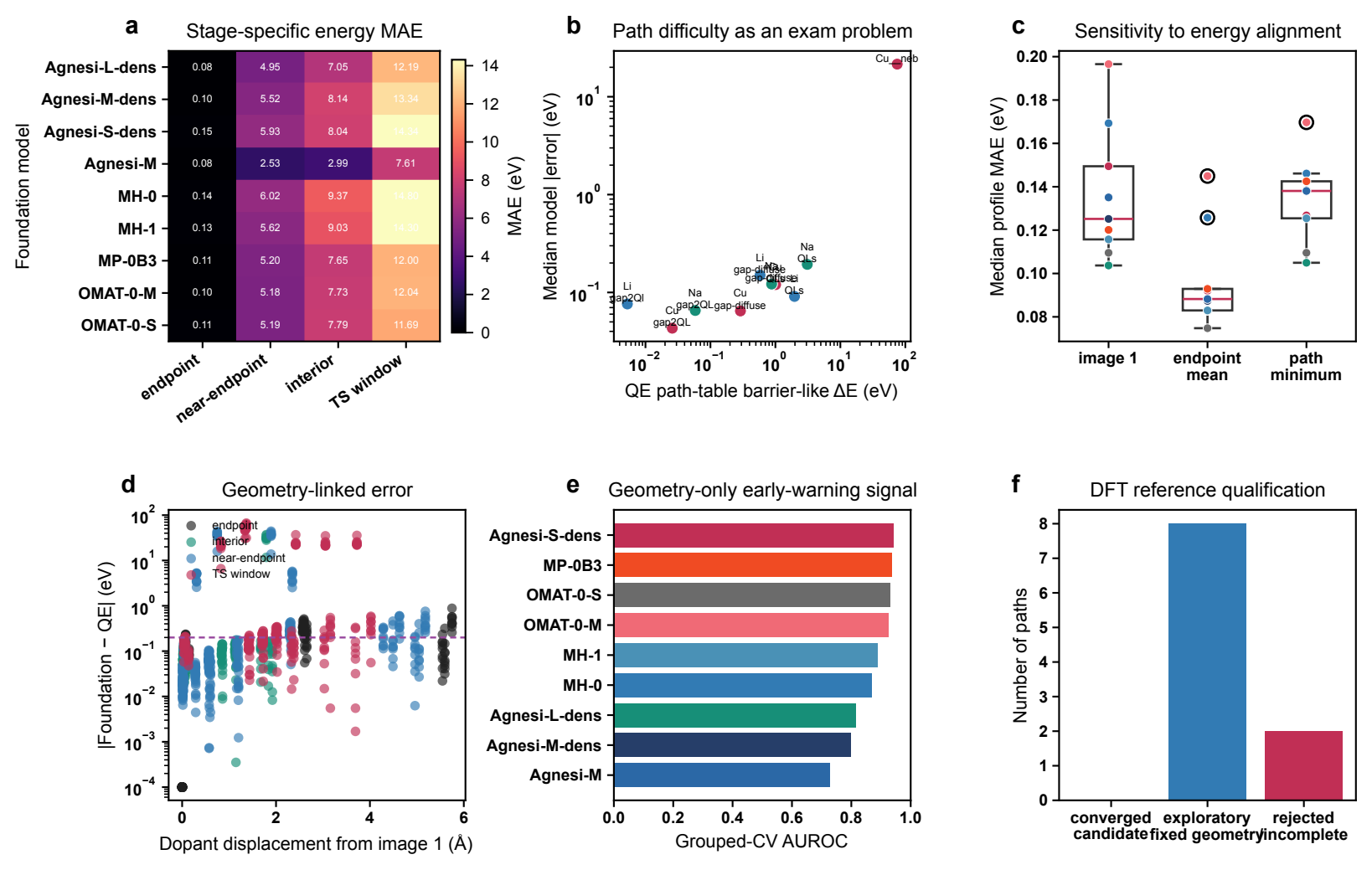

a
Stage-specific energy MAE
Foundation model
Agnesi-L-dens
Agnesi-M-dens
Agnesi-S-dens
Agnesi-M
MH-0
MH-1
MP-0B3
OMAT-0-M
OMAT-0-S
0.08 4.95 7.05 12.19
0.10 5.52 8.14 13.34
0.15 5.93 8.04 14.34
0.08 2.53 2.99 7.61
0.14 6.02 9.37 14.60
0.13 5.62 9.03 14.30
0.11 5.20 7.65 12.00
0.10 5.18 7.73 12.04
0.11 5.19 7.79 11.69
endpoint
near-endpoint
interior
TS window
MAE (eV)
0
2
4
6
8
10
12
14
b
Path difficulty as an exam problem
Median model |error| (eV)
QE path-table barrier-like ΔE (eV)
Cu_neb
Li gap2QL
Na gap2QL
Cu gap2QL
Cu gap-diffuse
Li QLs
Na QLs
c
Sensitivity to energy alignment
Median profile MAE (eV)
0.08
0.10
0.12
0.14
0.16
0.18
0.20
image 1
endpoint mean
path minimum
d
Geometry-linked error
|Foundation − QE| (eV)
endpoint
interior
near-endpoint
TS window
Dopant displacement from image 1 (Å)
e
Geometry-only early-warning signal
Agnesi-S-dens
MP-0B3
OMAT-0-S
OMAT-0-M
MH-1
MH-0
Agnesi-L-dens
Agnesi-M-dens
Agnesi-M
Grouped-CV AUROC
f
DFT reference qualification
Number of paths
converged candidate
exploratory fixed geometry
rejected incomplete